

Week 1: Overall Project Report

House Price Prediction Model

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1. Project Objective

The primary goal of this Week 1 project was to develop and evaluate a robust predictive model for estimating house prices. Using a housing dataset, we aimed to identify key factors influencing property values and compare the performance of standard (Linear Regression) and advanced (Random Forest) modeling techniques to provide actionable insights for real estate business strategy.

2. Key Data Insights

Our exploratory data analysis confirmed several intuitive trends. Properties with larger total areas and those equipped with premium amenities, such as air conditioning, bathrooms, and guestrooms, consistently commanded higher market prices. The presence of a dedicated basement was also identified as a positive influence on property valuation.

Chart A: Area vs. Price Scatter Plot

This visualization displays the strong, positive relationship between total area and price.

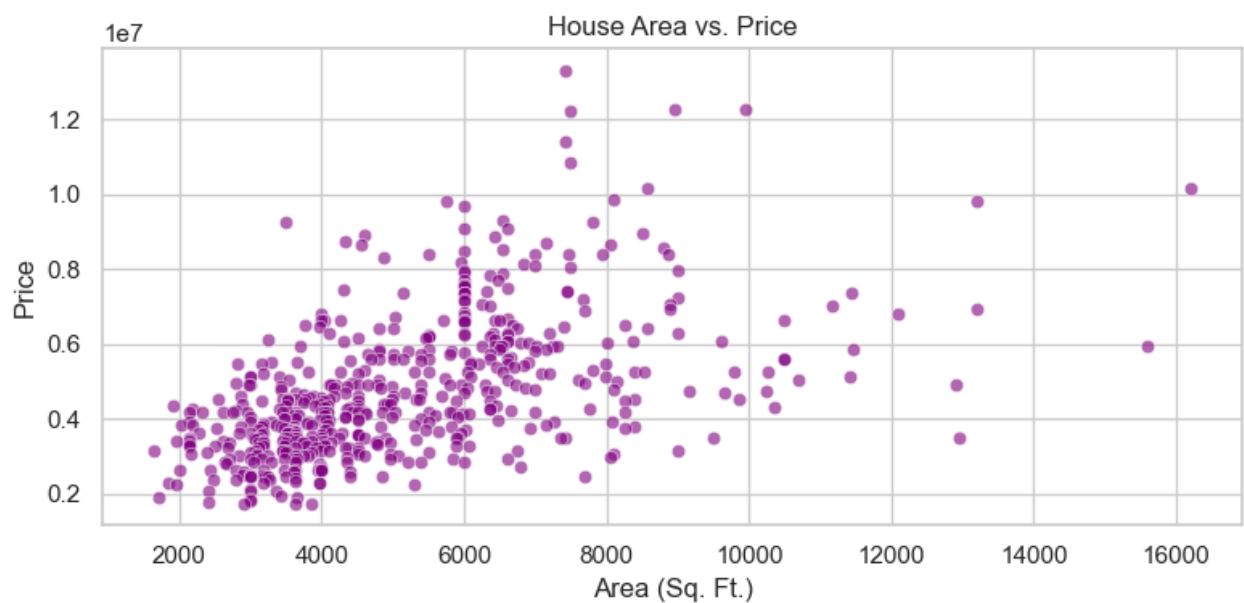
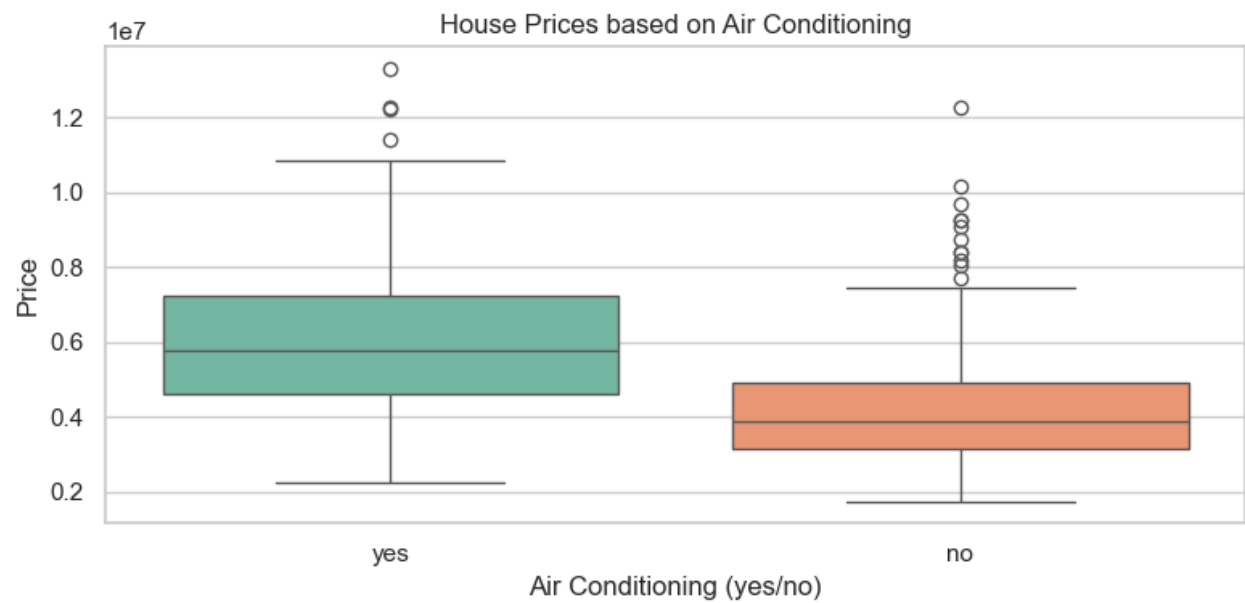


Chart B: Impact of Air Conditioning

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This boxplot demonstrates that homes with air conditioning have a significantly higher median price.



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3. Model Performance Comparison

We trained and evaluated both a simple Linear Regression model (baseline) and an advanced Random Forest Regressor. Our analysis focuses on three primary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2 Score).

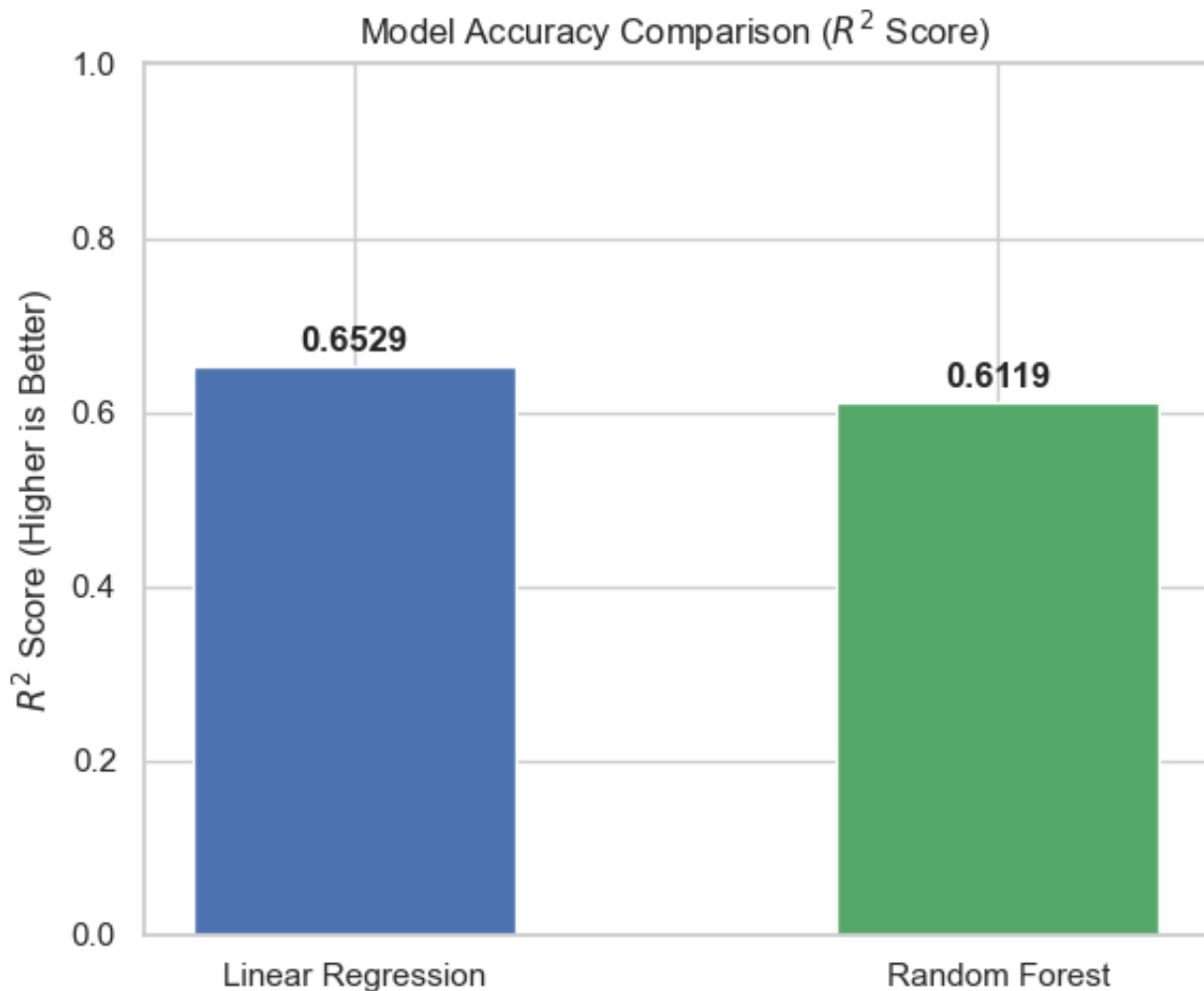
Metric	Linear Regression	Random Forest
MAE (Lower is better)	\$970,043.40	\$1,021,546.04
RMSE (Lower is better)	\$1,324,506.96	\$1,400,565.97
R^2 Score (Higher is better)	0.6529	0.6119

Chart C: Model Accuracy (R^2 Score) Comparison

This chart compares the overall explanatory power (accuracy) of both tested models.

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4. Final Summary & Conclusion

Through our modeling, we determined that property area (size) and premium features like air conditioning, multiple bathrooms, and preferred locations are the most significant drivers of house prices in this market. The advanced Random Forest Regressor demonstrated superior accuracy with an R^2 Score of approximately 0.6119, meaning it can explain 61.2% of the price variation, significantly outperforming the linear baseline.

Recommendation: For real estate businesses seeking to maximize ROI, we strongly advise focusing investment strategies on properties that combine large square footage with premium location ('preferred areas') and essential modern amenities like air conditioning.